When Substances Change (p204-209)

Most changes that occur can be classed as physical or chemical changes depending on whether any new substances formed.

Physical Change

Identifying Physical Change

1. What is the key feature of a physical change?

2. List **four** types of physical change.
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   -
   -

3. Why are physical changes often reversible?

Change in Shape or Form

1. A change in shape can occur when a force is applied to _____ , _____ , _____, _____ , _____ or _____ the object.

2. Why are these changes classed as physical changes?

3. Give two examples of physical changes that involve a change of shape or form.
   i.

   ii.

Expansion and Contraction

1. Another type of physical change occurs when the _____ of a substance _____ or _____ .

2. Describe what happens when an object is heated.

3. Describe what happens when an object is cooled.

4. Explain how thermometers use the expansion and contraction of a liquid to measure temperature.

5. Explain how a hot-air balloon is able to lift off the ground.

Changes of State

1. What happens when enough heat is applied to an object?

2. What happens when an object is cooled enough?

3. What is melting?

4. Give an example of a substance melting.

5. What is freezing?

6. Give an example of a substance freezing.

7. What is evaporation?

8. Give an example of evaporation.

9. How is evaporation different to boiling?

10. What is condensation?

11. Give an example of condensation?

12. What is sublimation?

13. Give an example of a substance that sublimes and what it is commonly used for.

14. Why are changes of state classed as physical changes?

Mixing

1. Give an example of a mixture that can be seen with the naked eye.

2. What happens when a solid (solute) is dissolved in a liquid (solvent)?

3. Describe how you can get the solute back out of the solution.

4. Describe an example of a solution.

5. Why is mixing an example of a physical change?

Chemical Change

Identifying Chemical Change

1. How do you know if a chemical change has occurred?

2. Give four ways you can detect the presence of a new substance.
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   -
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Permanent Colour Change

1. How is a colour change an indication of a chemical change?

2. Give four examples of chemical changes that involve a change in colour.
   i.
   ii.
   iii.
   iv.

3. Why is mixing paint classed as a physical change even though it causes a colour change?

Production of Gases

1. How can gas production be observed during a chemical reaction?

2. Give an example of a chemical reaction that produces carbon dioxide.

3. Why is rotting an example of a chemical change?

4. What is an example of a physical change that produces gases?

Forming a Precipitate

1. Explain what happens during a precipitation reaction.

2. Give two examples of precipitation reactions in everyday life.
   i.

   ii.

Energy is Produced or Absorbed

1. Energy can be produced or absorbed in the form of _____ , _____ or _____ .

2. A chemical change that produces energy is called an _____ reaction and a chemical change that absorbs energy is called an _____ reaction.

3. Give two examples of exothermic chemical changes.
   i.

   ii.

4. Give two examples of endothermic chemical changes.
   i.

   ii.

5. Give an example of a physical change that is endothermic.

6. Give an example of a physical change that is exothermic.